

Boundary-Induced Collapse:

A Realist Interpretation of Quantum Measurement from the TNA Framework

Abstract

Quantum mechanics is commonly interpreted as fundamentally indeterministic, with randomness treated as an ontological property of nature. In this work, we propose an alternative realist interpretation in which quantum randomness is not intrinsic, but emerges from incomplete access to boundary conditions governing the projection of higher-order structural states into observable configurations. Within the TNA framework, quantum measurement is modeled as a boundary-induced phase transition rather than a stochastic collapse. The wavefunction collapse is reinterpreted as a deterministic structural reconfiguration driven by a boundary shift. This approach naturally accounts for non-local correlations, the failure of local hidden variables, and the statistical success of quantum mechanics without invoking ontological randomness. Quantum indeterminacy is therefore interpreted as epistemic rather than ontological, arising from hidden boundary operators mediating the projection from $\aleph_1$ to $\aleph_0$.

Keywords

Quantum measurement; boundary conditions; wavefunction collapse; TNA; aleph hierarchy; non-locality; realism; phase transition; morphogenesis.

1. Introduction: The Interpretational Crisis in Quantum Mechanics

Quantum mechanics combines unparalleled predictive power with persistent ontological ambiguity. While its mathematical formalism remains uncontested, the interpretation of measurement and randomness continues to divide foundational physics.

The Copenhagen interpretation asserts that quantum outcomes are fundamentally random and that wavefunction collapse is irreducibly probabilistic. This position elevates epistemic limitation into an ontological principle.

However, this stance faces enduring conceptual difficulties:

- Absence of a physical collapse mechanism.
- Observer dependence.
- Non-local correlations incompatible with classical causality.

We argue that the core error lies in confusing *ignorance of boundary structure* with *ontological randomness*.

2. The End of Ontological Randomness

Randomness is not a property of reality itself, but a symptom of structural blindness.

In morphogenetic systems, abrupt transitions appear stochastic when boundary conditions are unknown. Once boundaries are specified, transitions become fully constrained. The same logic applies to quantum systems.

Thus:

Quantum randomness is epistemic rather than ontological.

Wavefunction collapse is not a random selection among possibilities, but a boundary-induced phase transition selecting the only configuration compatible with the new boundary.

3. Boundary Hidden Variables

Einstein searched for hidden variables inside particles. This search repeatedly failed.

We propose that determinism is not internal but *boundary-based*.

Let $\aleph_1$ denote the structural domain of latent configurations and $\aleph_0$ the observable projection. The transition

$$\aleph_1 \xrightarrow{B} \aleph_0$$

is governed by boundary operators B that are not fully accessible to observation.

Consequently:

- Local hidden variables fail (Bell).
- Statistical models succeed.
- Internal determinism remains undetectable.

Because determinism resides in the boundary mapping, not in the particle.

Bell's theorem excludes local internal hidden variables (Bell, 1964), but not global or boundary-based determinism. This aligns with Bohmian mechanics (Bohm, 1952) and modern superdeterministic approaches ('t Hooft, 2016; Hossenfelder & Palmer, 2020), while extending them through an explicit boundary formalism.

4. Measurement as Boundary-Induced Structural Transition

Measurement is not passive observation. It is an active redefinition of boundary conditions.

Let $B(t)$ denote the effective boundary configuration. Measurement induces

$$B(t) \rightarrow B(t + \Delta t),$$

destabilizing the superposed configuration and forcing reorganization toward the unique state minimizing the structural functional $F_{\min}$.

Collapse is not choice.

It is structural necessity.

Quantum measurement is therefore a boundary-induced phase transition rather than a stochastic event.

This interpretation is consistent with measurement-induced entanglement phase transitions (Skinner, Ruhman & Nahum, 2019; Li, Chen & Fisher, 2019).

5. Biological Parallel: Phase Transitions under Boundary Shifts

Biological systems exhibit structurally analogous behavior.

Biomolecular condensates undergo liquid–liquid phase separation when boundary parameters such as concentration, temperature, or pH cross critical thresholds (Brangwynne et al., 2009; Hyman, Weber & Jülicher, 2014).

Developing tissues experience rigidity and fluidization transitions governed by boundary-controlled mechanical constraints (Manning et al., 2010; Merkel & Manning, 2018).

These transitions demonstrate that abrupt morphogenetic reorganizations induced by boundary changes are both computationally and physically realizable.

6. Non-Locality Reinterpreted

Bell's theorem excludes local hidden variables, not global boundary operators.

Entangled particles do not communicate.

They respond to the same boundary structure.

Their correlations are projections of shared boundary constraints.

Non-locality therefore requires no superluminal signaling.

7. Formal Representation

Boundary field: $B(t)$

Projection operator: $\Pi(\psi|B)$

Structural mapping:

$$\aleph_1 \xrightarrow{B} \aleph_0$$

Observed probabilities:

$$P(\psi) = \Pi(\psi|B_{\text{unknown}})$$

Randomness arises from incomplete boundary knowledge.

8. Formal Boundary Projection Expression

$$\Pi(\psi|B) : \aleph_1 \xrightarrow{B} \aleph_0$$

Measurement:

$$B(t) \rightarrow B'(t)$$

forcing a discontinuous projection transition.

9. Distinctive Predictions

1. Apparent randomness decreases with boundary knowledge.
 2. Non-local correlations require no signaling.
 3. Collapse dynamics are environment-dependent.
 4. Critical thresholds should appear in measurement statistics.
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10. Connection with Macroscopic Transitions

The same structural law applies to:

- Chernobyl ecological reorganization.
- Cambrian explosion.
- Cancer progression.

Boundary shifts induce structural reorganization, not stochastic drift.

11. Philosophical Closure

The universe is not governed by dice.

It is governed by hidden boundaries.

God does not play dice.

God hides the cylinders.

(Ontological metaphor, not theology.)

12. Conclusion

Quantum mechanics does not require ontological randomness.

Wavefunction collapse is a boundary-induced phase transition.

Quantum indeterminacy is not chaos, but hidden structure.

Final Sentence

Quantum indeterminacy is not the signature of chance, but the shadow cast by unseen boundaries.

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